

Orchid

Orchid Operation Manual

Version 4.1 | March 2026

1. Introduction

Welcome to the world of limitless chord creation. This is Orchid. An amazing new instrument that's changing the face of music making and production for artists all over the globe.

1.1 Thank You for Purchasing an Orchid

We sincerely appreciate your support and trust in Orchid. This instrument was designed with passion and innovation, and we hope it becomes an integral part of your creative journey. Whether you're using it for inspiration, performance, or exploration, we are thrilled to have you as part of our community. Your feedback and engagement help shape the future of Orchid, and we can't wait to hear what you create. Thank you for being a part of this adventure.

1.2 Overview: What Orchid Is and What It's For

Orchid is a chord-generating digital synthesizer, built on a revolutionary chord logic algorithm and industry-leading, voice-leading technology. It boasts three separate and powerful synth engines, a suite of on-board effects, various performance modes, killer real-time looping controls, and a funky beat machine preloaded with a selection of modern grooves.

Orchid's physical interface is so sophisticated, it can generate multi-layered, polyphonic ideas and upload them directly to your digital audio workstation. And yet, it's so simple, even a child can use it.

2. What's in the Box

2.1 Orchid



Looking at Orchid, you'll see:

- Two stereo speakers on the top of the case
- Nine endless rotary encoders along the front panel
- An OLED display
- Eight buttons on the left-hand side (four **Chord Type** buttons on top, four **Chord Extension** buttons below)
- A single-octave velocity-sensitive keyboard on the right-hand side
- A **Chord Voicing Dial** in the middle and a **Bass Voicing Dial** below it.

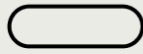
2.2 Included Accessories

- Quick Start Guide
- Notebook
- No charger (for environmental reasons, use any standard USB-C charger). We recommend using a fast charger.

3. Connections & Setup



MIDI



USB-C



Output

3.1 Audio Output

- **3.5mm Stereo Output Jack:** Connect headphones or an aux cable. Orchid's built-in speakers will automatically disable when this is in use (modifiable in **Options**).

3.2 MIDI Output

With either the **MIDI 5-Pin Out** or **USB-C MIDI Out** you can:

- Send chord, bass and performance notes on separate MIDI channels.
- Use Orchid as a MIDI controller for DAWs or external synths, or bi-directional control of Pistil.

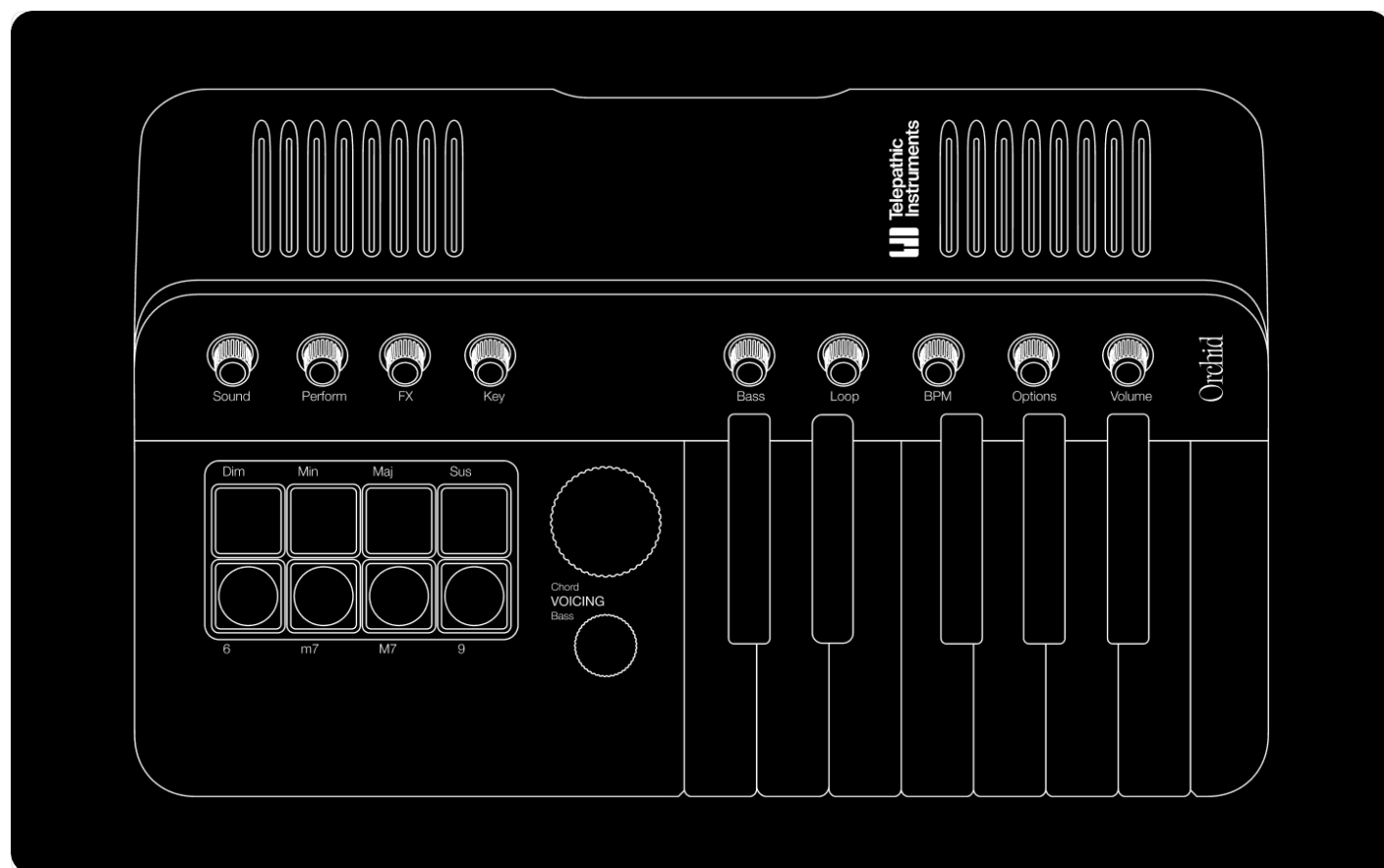
Default Channels:

- **Channel 1:** Performance
- **Channel 2:** Bass
- **Channel 3:** Chord

3.3 USB-C: Charging, MIDI & Software Updates

- **Charging:** Connect any USB-C charger. For maximum efficiency, plug Orchid into a wall socket.
 - **Note:** Orchid does not charge while in USB MIDI mode.
- **Software Updates:** See **Upgrade Firmware** in the **Options** section.

4. Getting Started: Orchid Tour



4.1 Powering On & Boot Sequence

To turn Orchid on, hold the power button for one second. When you turn on Orchid, you'll see the logo and a short boot sequence. Once this has completed, your Orchid is ready to play.

4.2 Adjusting Volume and Checking Battery

- **Master Volume:** Turn the **Volume Dial** (range 0–99).
- **Bass Volume:** Press and turn the **Volume Dial** to adjust bass volume independently.
- **Beats and Metronome Volume:** Press and turn the **BPM Dial** to independently adjust the volume of **Beats** or the **Metronome**.
- The LED next to the **Volume Dial** also indicates battery status, flashing with increasing speed as the charge decreases.
- **Quick Battery Check:** Push the **Volume Dial** to see the battery percentage.



5. Sound Selection & Chords

5.1 Sound Selection

Orchid's three polyphonic synthesizer engines drive over 50 onboard **Sounds**, ranging from classic vintage organs, to lush journeys through the Fangorn Forest, to Kilimanjaro, rising like Olympus above the Serengeti.

- Rotate the **Sound Dial** to browse through a variety of sound profiles by number.
- Press the **Sound Dial** then rotate to browse through sounds by name.
- Each sound combines synth presets, effects, and performance settings.
- Experiment with different sounds to find what inspires you.
- You can customise **Sounds** and save them for later (see 8. Effects & Sound Processing)



5.2 Playing Chords

With Orchid, making chords has never been easier!



- **Manual Chord Selection:**
 - Hold down a **Chord Type** button with your left hand (e.g., Dim, Min, Maj, Sus).
 - Press a note on the **Keyboard** with your right hand to play the corresponding chord.
- **Adding Complexity:**
 - Hold a **Chord Extension** button, along with a **Chord Type** button, before pressing a note on the Keyboard.
 - **Example:** Hold **Maj** and **M7** buttons, then press **C** to play a CMaj7 chord.

- Each **Chord Extension** adds new layers to the base chord, expanding harmonic possibilities.
- Try adding multiple extensions at once for complex chord voicings.
- Watch out for too much Jazz.
- Note that **Extensions** only work in addition to a **Chord Type**. You can't play **Extensions** by themselves.

5.3 Chord Mood Swings

Each **Chord Type** represents a different mood, influencing the harmonic character of the music:

- **Maj (Major)** – Bright, happy, resolved.
- **Min (Minor)** – Dark, melancholic, introspective.
- **Sus (Suspended)** – Unstable, open-ended, leading to resolution.
- **Dim (Diminished)** – Tense, dramatic, and unsettling.

Think of **Chord Types** as the main course and **Chord Extensions** as the seasoning that enhances the harmonic flavor.

5.4 Common Chord Combinations

Here's a few common combinations to get you started.

- **Major 7th (Maj7)**
 - **Hold [Maj] + [M7] Buttons**
 - Feels jazzy or fancy. Turns a happy Major chord into something with more class and style. It adds a slightly mysterious feel, without the darkness of a Minor chord.
- **Minor 7th (min7)**
 - **Hold [Min] + [m7] Buttons**
 - Softens the melancholic nature of the Minor chord, adding emotional depth and a soulful character.
- **Minor Major 7th (minM7)**
 - **Hold [Min] + [M7] Buttons**
 - Haunting and intense. It blends the darkness of a Minor chord with the brightness of a Major, creating a striking contrast.
- **Dominant 7th Suspended 4 (Sus4-7)**
 - **Hold [Sus] + [m7] Buttons**
 - Highly dissonant. It creates tension that demands resolution. (Consider choosing a better Sus variation.)
- **Diminished 7th (Dim7)**
 - **Hold [Dim] + [m7] Buttons**
 - Packed with tension, creating an anxious, unresolved sound. Great for generating unease while remaining in key.
- **Dominant 7th ([Note]7)**

- **Hold [Maj] + [m7] Buttons**
- Has a strong pull towards resolution. It leans towards being resolved by the root chord of the key which is always 7 semi-tones down (or 5 up) on the keyboard (e.g., G7 resolves to CMaj). This creates a powerful transition where tension builds and then releases.

Tip: There are two approaches to finding the perfect chord with extensions:

1. **The additive approach:** Start with a simple chord type and gradually add extensions one by one to build complexity.
2. **The subtractive approach:** Begin with all four **Chord Extension** buttons active (creating maximum complexity) and gradually remove them one by one. Listen carefully as you subtract each extension - can you identify which one is holding you back from the sound you're seeking? Just like a sculptor chisels away at stone, sometimes starting with more and refining through subtraction can provide just the perspective you need to discover the perfect chord.

Both approaches offer different creative pathways to chord exploration, and switching between them can help overcome creative blocks.

5.5 Chord Voicing

The **Chord Voicing Dial** allows you to adjust the harmonic structure of the chord by rearranging the note order. It allows you to move fluidly up and down the keyboard, one note at a time, causing a continuous cascade of chord voicing inversions, dramatically altering a chord's tonal character.

Voicings work by taking the lowest note currently being played in the chord and moving it up an octave to the top, and vice versa. This creates different inversions of the same chord while maintaining its harmonic function. This allows you to create smooth, elegant voice-leading between chords in a progression, avoiding the clunky, static jumps that can occur when all chords are played in their root position.

- Turning **up** moves the lowest note **up an octave**.
- Turning **down** moves the highest note **down an octave**.

Press the **Chord Voicing** dial once to switch between **Split** and **Octave** mode. **Split** mode places the octave jump at a point that you choose on the keyboard, so you'll notice that your notes suddenly jump up or down an octave. **Octave** mode means that the entire keyboard will stay within the octave that you choose.

Tip: Experiment with chord voicing to make your progressions more interesting and avoid static-sounding harmonic movement.

6. Orchid Chord Language

There are lots of different ways to talk about chords:

Chord Type	Notation Examples
Major	C, CM, Cmaj, CMaj
Minor	Cmi, Cm, C-
Major 7th	Cmaj7, CΔ7, CM7
Minor 7th	Cm7, Cmi7, C-7
Dominant 7th	C7, Cdom7
Diminished	Cdim, Co
Half-Diminished	Cø, Cm7 b 5
Augmented	C+, Caug

The diversity in notation usually reflects genre conventions:

Jazz prefers shorthand and compact symbols to minimize clutter.

- **Example:** C- (minor), Cmaj7 or CΔ7 (major seventh), Cø (half-diminished).
- **Pros:** Compact and widely recognized in jazz contexts, allowing quick reading.
- **Cons:** Can be confusing for those unfamiliar with jazz-specific symbols (e.g., Δ for major seventh).

Classical theory offers precision but often at the expense of being verbose and impractical. Like it's been written by hand. In the 17th century.

- **Example:** C minor, C minor 7th.
- **Pros:** Precise and clear, ideal for teaching and formal scores.
- **Cons:** Verbose, not practical for real-time performance or software interfaces.

Digital music platforms and Digital Audio Workstations (DAW) balance simplicity and readability for ease of use.

- **Example:** C, Cm, C7, CM7,
- **Pros:** Balances simplicity and readability, accessible for users across skill levels.
- **Cons:** Lacks the extreme shorthand used in jazz but supports cross-genre accessibility.

6.1 Chord Notation: Orchid Standard Framework

The Orchid Standard Chord Naming Framework is our first attempt to provide a clear and consistent system for chord notation. This framework focuses on simplicity and ease of use, and so draws more heavily on Jazz notation with a DAW-ish approach.

Our goals are:

- **Ease of Use:** An intuitive naming system for musicians of all levels.

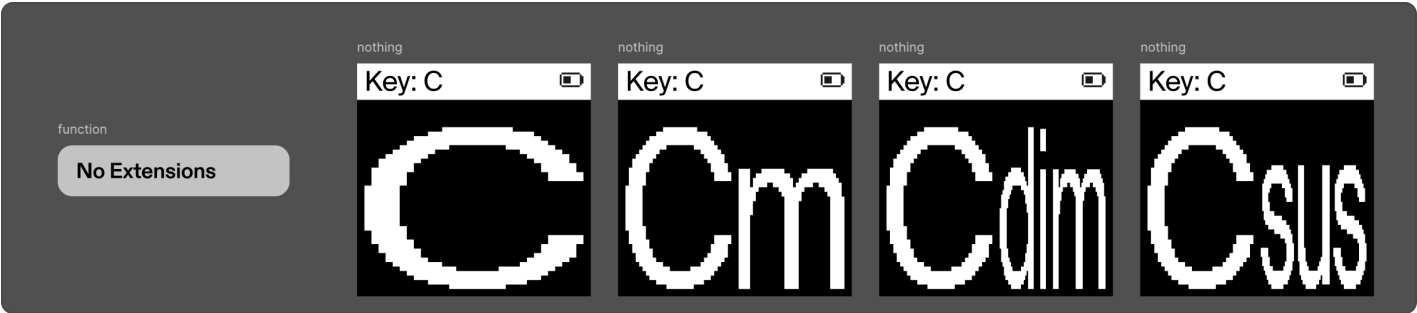
- **Compatibility:** Seamlessly integrates with DAWs and composition software.
- **Scalability:** Future modular modes ensure adaptability for advanced users.
- **The Right Amount of Nerd:** Which is to say some amount of nerd.
- **Efficiency:** Needs to fit onto a tiny-ass screen.

6.2 Triads (Top Keys)

The top keys show what type of chord we’re dealing with, and are in the same size font as the root note of the chord (e.g. Cm).

6.3 Orchid Standard Framework: No Ext

Chord Type	Extension	EG
Major	None	C
Minor	None	Cm
Suspended	None	Csus
Diminished	None	Cdim



6.4 Extension Rules

Extensions are handled as little letters and numbers at the top right (like you slept through in maths class). To keep things simple (and because of the way Orchid’s voicing engine works) we don’t do fancy stuff like turn a “6” into a “13”. What you play is what you get. And to keep things simpler still, anything that gets too busy (i.e. too many extensions) will turn into a JAZZ chord. (If you want to see what those notes are, you can turn on Geek Out mode.)

The only real anomaly to this rule is the Major chord with a minor 7th (dominant), which is very typically just written as C⁷.



1. **One Extension:**
 - Extension is in superscript, e.g., **CM⁷** (C Major 7th).
2. **Two Extensions:**
 - Combine both in superscript, e.g., **CM⁷⁹** (C Major 7th + 9th).

3. Three Extensions ("JAZZ"):

- Use **JAZZ** when there are three extensions, e.g., **Cm^{JAZZ}**.

4. Four Extensions ("WTF"):

- Use **WTF** for four extensions, e.g., **CWTF**.

6.5 Orchid Standard Framework: Extensions

Top Keys	Bottom Keys	EG
Major	None	C
	6th	C ⁶
	Major 7th	C ^{M7}
	Minor 7th	C ⁷
	9th	C ⁹
	Major 7th 9th	C ^{M79}
	Major 7th 6th	C ^{M76}
Minor	None	Cm
	6th	Cm ⁶
	Minor 7th	Cm ⁷
	Major 7th	Cm ^{M7}
	9th	Cm ⁹
	Minor 7th 6th	Cm ⁷⁶
	Minor 7th 9th	Cm ⁷⁹
	Major 7th Minor 7th 9th	Cm ^{JAZZ}
Suspended	None	Csus
	6th	Csus ⁶
	Minor 7th	Csus ^{m7}
	Major 7th	Csus ^{M7}
	9th	Csus ⁹
	Minor 7th 6th	Csus ^{m76}
	Major 7th 9th	Csus ^{M79}
	6th Minor 7th 9th	Csus ^{JAZZ}
Diminished	None	Cdim
	6th	Cdim ⁶

Minor 7th	Cdim ⁷
9th	Cdim ⁹
6th Minor 7th	Cdim ^{Jazz}
9th	
Major 7th 9th	Cdim ^{WTF}
Minor 7th 6th	

7. Performance Modes

A sustained chord is nice, but it can get a bit boring. That's why Orchid has over three mind-blowing and configurable **Performance Modes**, and definitely no fewer than 10 pre-set melodo-rhythmic **Patterns**, all with variable BPM control.

7.1 Accessing Performance Modes

- Press the **Perform Dial** to see the available modes.
- Turn the **Perform Dial** to scroll through the modes.
- Press the **Perform Dial** to select a mode.
- Turn the dial again to adjust the selected selected mode.



7.2 Available Performance Modes

Strum

- Mimics guitar strumming by staggering chord notes from low to high.
- Adjust the speed using the **Perform Dial**.
- Creates a natural, flowing effect similar to stringed instruments.

Strum 2 Octaves

- Similar to Strum but extends the range to two octaves for a wider, more dramatic effect.

Slop

- Similar to Strum, but with timing imperfections for a more human feel.
- Adjust the level of Slop to introduce randomness in note timing for a natural, humanized sound.

Arpeggiate

- Breaks chords into sequences of notes played in order.
- Plays the notes of your chord in a repeated sequence selected by turning the **Perform Dial**, using 1/8 or 1/16 notes.
- Every note in the chord will be voiced; a DMaj will voice 3 notes (D, F#, A), while a DMaj7 will voice 4 (D, F#, A, C#).
- The more notes/extensions you select, the longer the arpeggiator sequence will be.

Arpeggiator 2 Octaves

- Extends the arpeggiator range across two octaves for more dramatic patterns.

Pattern

- Plays chord notes in pre-determined rhythmic patterns.
- Unlike the **Arpeggiator**, Pattern mode plays notes in sequences that Orchid has selected for you.
- Keeps notes in time with the music tempo, even if the number of notes in the chord changes.
- **Patterns** contain velocity information too. Notes in the **Pattern** will play at different volumes, scaled to how hard you play the keyboard note.

Harp

- Plays the succession of notes in the chord up many octaves fast, like a harp.
- Creates sweeping, expansive passages ideal for dramatic moments.

All timing-based performance modes sync with the **BPM**, allowing you to speed up or slow down the sequence as needed.

Tip: Experiment by combining different **Performance Modes** with **Effects** for truly unique soundscapes!

7.3 Locking Performance Settings

Long-press the **Perform Dial** to lock the current performance settings. When performance lock is active, an LED beside the **Perform Dial** will illuminate.

Locking the settings means the effects will stay locked in place as you explore new sounds with the **Sound Dial**. This ensures that even when you change **Sound**, your performance settings remain unchanged, maintaining consistency in your sound design.

8. Effects & Sound Processing

Add your final layer with the onboard effects processor. Orchid has a range of built-in sound effects designed to enhance and shape your sound. These effects allow you to create anything from subtle warmth to dramatic soundscapes.

8.1 Available Effects

- **Reverb:** Simulates the lushness of sending your notes into a large space. Whether you want your keyboard to sound like it's being played in a cathedral or a small room, Reverb adds depth and spatial character to your sound.
- **Delay:** Creates repetitions of everything you play while gently fading the resulting copies away (commonly called echo). Adjust delay time and feedback for rhythmic complexity.
- **Chorus:** Duplicates the sound and slightly detunes the copies. This brings more depth and fullness, making a single note feel like it's being played by multiple instruments.
- **Phaser:** Produces a classic swirling effect, shifting frequencies dynamically to create a sweeping, psychedelic tone. Often associated with the 70s rock and electronic music scene.
- **Flanger:** Takes a duplicate of the signal and moves it closer and further away (in time) to the original, creating a distinct, artificial-like flavor that sounds metallic or jet-engine-like.
- **Drive:** Adds subtle warmth or aggressive distortion, giving the sound more character and grit. Perfect for thickening tones and making them stand out.
- **Tremolo:** Modulates the volume up and down in a rhythmic pattern, creating a pulsating effect that can range from subtle movement to dramatic amplitude shifts.
- **Ensemble:** Expands on the Chorus effect by introducing multiple layers of detuned voices, mimicking the sound of a string ensemble or multi-voice synth.
- **Filter:** Allows you to sweep through the frequencies of the main synth engine for dynamic, expressive filter effects.
- **Drum FX Reverb:** Adds depth and spatial character to Beats.
- **Drum FX Saturation:** A subtle form of harmonic distortion that adds richness, warmth, and character to Beats by creating new overtones and performing a gentle form of compression.

Exit	
Reverb	05
Chorus	02
Delay	01

8.2 Adjusting Effects

Each programmed sound in Orchid has a combination of FX stored within it. By default, the **FX Dial** will automatically increase and decrease the most prominent effect of the currently selected sound.

To customize the effects:

- Push the **FX Dial** to see the available effects.
- Turn the **FX Dial** to choose the effect you wish to change.
- Push the **FX Dial** to select the effect.
- Turn the dial to adjust the selected effect's intensity.



8.3 Locking FX Settings

Long-press the **FX Dial** to lock the current FX settings. When FX lock is active, an LED beside the **FX Dial** will illuminate.

Locking the settings means the effects will stay locked in place as you explore new sounds with the **Sound Dial**. This ensures that even when you change **Sound**, your effect settings remain unchanged, maintaining consistency in your sound design.

8.4 Saving User Sounds

You can save customized sounds as **User Sounds** to access later. This captures your custom **Performance** and **FX** settings.

- Long press the **Sound Dial** to enter the **Sound Save** menu.
- Save your **Sound** as the suggested **User Sound** number or push **Save As** to pick the number you want.
- Your **User Sound** will now appear in the **Sound** menu.
- Long press the **Sound Dial** then cycle to **Delete** to the currently selected **User Sound**.

9. Key Mode & Transposition

With **Key Mode** activated, any note you play on the keyboard will automatically generate a chord that fits within that key. This frees you from the need to think about complex music theory while you are using Orchid.

9.1 Accessing Key Mode

- Press the **Key Dial** to enable or disable **Key Mode**.
- Turn the **Key Dial** to select your desired key.

Note: when **Key Mode** is enabled, your selected key will display on the screen, and an LED light will show next to the **Key Knob**.

9.2 Quick Key Select

- Long press the **Key Dial** until “select key” and keyboard appears on the display.
- Press the root note of the desired key on the keyboard to select that key.
- For a minor key, hold the minor **Chord Type** button when pressing the keyboard note.

9.3 Key Mode Functionality

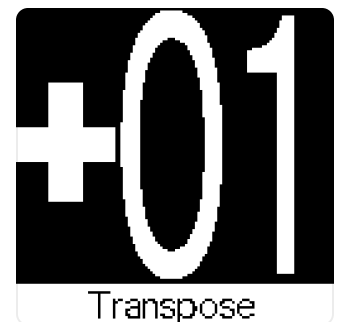
When turned on, **Key Mode** automatically generates chords based on the selected key.

- **Example:** If you select **C Major** and lock it in, the right-hand keyboard will trigger chords using only notes from the **C Major scale** (all white keys in the case of C Major).
- Ensures that all played notes remain harmonically correct, eliminating the need to manually select chord voicings.
- If you play an out of key note while in **Key Mode**, Orchid will intelligently shift the note to the nearest note within the scale, depending on the context. For example, if you press C# while in the key of C, Orchid will play a Csus.
- You can override key mode temporarily by playing chords as usual using the **Chord Type** buttons.



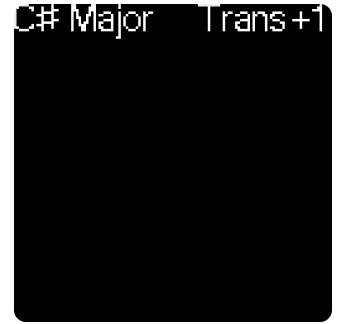
9.4 Transposition

- Press and turn the **Key Dial** to transpose the entire output of Orchid by a semi-tone at a time.
- This is useful for adjusting the output to match vocalists or other musicians.



9.5 Visual Indications

- When **Key Mode** is active, the selected key will be displayed in the **top-left corner** of the screen.
- When **Transposition** is active, the transposition amount is shown in the **top-right corner**.



Tip: In **Key Mode**, each note you play triggers a chord using only notes that fit within the selected key. Thanks to **harmonic quantization**, any out-of-scale notes are automatically adjusted, ensuring seamless harmonic consistency.

10. Bass Sound & Voicing

Adding a layer of **thundering bottom end** to your chords can create depth and richness in your sound. Orchid has a dedicated **bass synth engine** that operates independently of the **main synth engine**.

10.1 Activating Bass Mode

- Press the **Bass Dial** to toggle **Bass Mode** on/off.
- When active, **Bass Mode** adds the **root note** of the chord through a dedicated bass synth engine.
- Turn the **Bass Dial** to cycle through a range of sexy bass sounds.

10.2 Adjusting Bass Behavior

By holding down the **Bass Dial** for one second, you can access a menu to adjust the bass mode behavior:

- **Chords Only:** Bass notes play only when chords are played.
- **Unison Bass:** Bass notes match exactly what is played on the keyboard.
- **Single Notes:** Bass notes sound with every keyboard note pressed, while the main synth engine only sounds when a chord is played.
- **Solo:** The bass plays independently without other elements.

10.3 Bass Voicing

Similar to **Chord Voicing**, the **Bass Voicing Dial** allows you to modify the way bass notes are played:

- Moving the voicing **down** results in lower, deeper bass notes.
- Moving the voicing **up** shifts the bass notes higher in pitch.

10.4 Adjusting Bass Volume

- The **Bass Volume** is adjusted separately from the main volume.
- Hold the **Volume Dial** while turning it to adjust bass volume independently.

11. BPM Control, Beats & Timing Features

Several features in Orchid rely on a master **BPM** setting to control the overall pace of time-based effects and performance modes. The **BPM Dial** serves as the global tempo control, ensuring everything stays in sync. And it's here that you can access the magic of **Beats**.

11.1 Adjusting BPM

- Turn the **BPM Dial** to set the tempo (beats per minute).
- The current **BPM value** is displayed on the screen for precise control.



11.2 Using the Metronome

- Push the **BPM Dial** once to turn the metronome on/off.
- Press and turn the **BPM Dial** to adjust the metronome volume.
- Hold the **BPM Dial** for one second then scroll to select a time signature.
- Change the **Metronome Sound** in the **Options** menu.

11.3 Synchronization & Timing

The **BPM Dial** ensures that all time-based features stay aligned:

- **Performance Modes:** Arpeggiator, Pattern, and Loop Mode sync to BPM.
- **Delay Effect:** Repeats will follow the BPM setting for rhythmic accuracy.
- **Loop Recording:** If a tempo-based loop length is selected, it will automatically align to BPM.

Tip: Crank up the **BPM** until the **Arpeggiator** blurs together. By creatively manipulating time, you can discover unique and unexpected sonic textures!

11.4 Beats

Orchid comes with a selection of pre-programmed **Beats**.

- Long press the **BPM Dial** for one second and scroll past the time signatures to access **Beats**.
- Press the **BPM Dial** to start or stop a **Beat**.
- Turn the **BPM Dial** to change the tempo of **Beats**.
- Press the **FX Dial** and scroll to access **Drum FX**, where you can customize reverb and saturation on your drum sound.
- Press and turn the **BPM Dial** to adjust the volume of the **Beat**.

12. Loop Mode & Layering

Loop Mode allows you to record, layer and manipulate musical phrases in real time. Whether you're laying down chords, melodies, or rhythms, **Loop Mode** helps you capture your ideas and build upon them step by step.

12.1 Entering Loop Mode

- Push or turn the **Loop Dial** to access the **Loop Mode Waiting Room**.
- Choose your preferred looping method from the available options.

12.2 Loop Mode Options

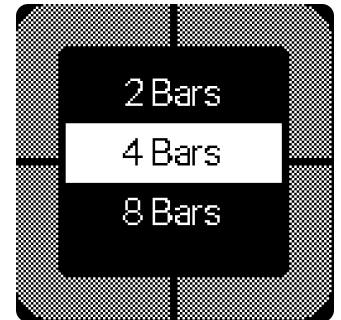
The **Loop Dial** functions differently depending on your current stage of looping:

- Turn the **Loop Dial** to select your **Sync Mode**:

- **Free Mode:** Record without tempo restraints, continuing until manually stopped.
- **BPM Mode :** Choose from **1 bar, 2 bars, 4 bars, 8 bars, or 16 bars** of recording at the current BPM.

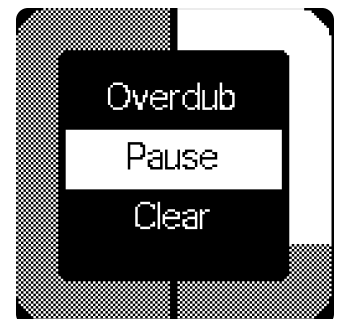
12.3 Recording a Loop

- Press the **Loop Dial** to **start recording** once you have selected your **Sync Mode**.
- In **BPM mode**, a **1-bar count-in** will play via the metronome before recording starts.
- In **Free mode**, recording starts as soon as you play the first note.
- The **display shows "Rec"** to indicate active recording.
- A **Ring Progress Indicator** on the screen provides a visual representation of the recording length.



12.4 Playback & Loop Control

- Press the **Loop Dial** on **Stop** to stop recording in **Free Mode**.
- Wait for the fixed loop to finish to stop recording in **BPM Mode**.
- While the loop is playing back:
 - Select **Pause** or **Play** to pause or resume playback.
 - Select **Clear** to **clear the loop** and return to the **Loop Mode Waiting Room**.
 - Select **Overdub** to continue recording a new layer.
 - Long press the **Bass Voicing Dial** and press the **Loop Dial** to instantly stop your **Loop** and leave **Loop Mode**.



12.5 Overdubbing & Undoing Layers

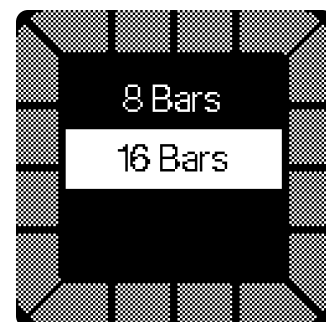
Overdubbing allows you to add layers to your existing loop.

- New notes will be recorded on top of the loop.

- Press the **Loop Dial** on **Stop** to finish overdubbing.
- If needed, remove the last layer you recorded by selecting **Undo** from the menu.
- If you undo all overdubs back to the original recording, the **Undo** option changes to **Clear**, which will **erase the loop completely**.

12.6 Metronome/Beats & Visual Indicators

- The **Metronome** or **Beats** can be toggled on/off by pressing the **BPM Dial**.
- Adjust the **Metronome/Beats Volume** by pressing and turning the **BPM Dial**.
- The **Ring Progress Indicator** around the edge of the display shows your current position in the loop, making it easier to time new recordings.



12.7 Save, Load and Delete Loops

You can save up to 10 **Loops** on Orchid.

- Long press the **Loop Dial** to access the **Save Loop** menu.
- Select **Save As Loop XX** to save your loop in the next available spot.
- Select **Save As** to select which spot to save your **Loop** in.
- Select **Load Loop** to retrieve and play your saved **Loops**.
- Select **Delete Loop** to delete your saved **Loops**.

Tip: Want more flexibility when building chords? Use **Loop Mode in Overdub** to record **one note of the chord at a time**. Each loop pass allows you to layer a new note, letting you experiment with complex chord voicings. If the latest note isn't right, simply **undo the action and try again**. This method gives you precise control over your chord stacking and timing, especially when adjusting with the **BPM Dial**.

14. Options Menu

The **Options Menu** contains a variety of settings that change how Orchid operates, as well as system information. To access these options, use the **Options Dial**.

14.1 Battery

- Displays the **current battery level**.
- At **10% battery**, a warning notification appears.

- At a critical level, Orchid will **automatically shut down**.
 - Plug into a wall charger to charge Orchid most efficiently in this state.
- The **LED next to the Volume Dial** will flash when battery is low, increasing in speed as the charge depletes.

14.2 View

Select different **View Modes** to customize the display:

- **React:** Displays an oscilloscope for a real-time visual representation of the waveform.
- **Chord:** Displays only the **current chord** being played in large text.
- **Keyboard:** Shows a visual keyboard with highlighted notes being played.
- **Chord & Keyboard:** Displays both the **keyboard** and the **chord name**.
- **Notes:** Displays the **chord name** and the **individual notes** being played in written format.
- **Geek Out:** Displays **maximum information**, including the keyboard, chord name, and notes.



14.3 Audio Output

Customize how audio is routed when an external device is connected:

- **Auto:** Automatically detects and adjusts audio output.
- **Headphones:** Outputs only through the **headphone jack**, disabling speakers.
- **Speakers:** Outputs only through **built-in speakers**, muting the headphone output.
- **Both:** Plays sound through **both** speakers and headphone output simultaneously.

14.4 MIDI Channels

Determine what MIDI channel Orchid sends MIDI note data on:

- **Performance:** Outputs **notes based on Performance Modes** (e.g., Arpeggiator, Strum).
- **Bass:** Sends **bass notes separately** on an independent MIDI channel.
- **Chord:** Outputs **played chords**, even if a Performance Mode is active.

All three channels can be used at once, allowing for **separate recording** in a DAW or **triggering multiple instruments**.

14.5 Play Style

Adjust Orchid's **Play Style** to match your preferred way of interacting with the instrument:

- **Simple** – Simplified controls for quick and intuitive play.
- **Advanced** – Provides more control and customization.
- **Free** – Full creative freedom with minimal restrictions.

14.6 Extension Addition

Controls how **Chord Extensions** behave when added:

- **Add Note** – Adds only the additional extension without retriggering the full chord.
- **Play Chord** – Replays the **full chord** when extensions are added.

14.7 Single Note Mode

- **Split Keyboard** – Allows splitting the keyboard into different playable zones.
- **Full Octave Keyboard** – Keeps the keyboard as a full playable octave.

14.8 Secret Chords

Turning on **Secret Chords** gives you access to a range of extra spicy chords:

Secret Chords

Top Keys	Bottom Keys	EG
Dim + Sus	None	C ⁵
Maj + Sus	None	C+
Min + Sus	None	Cm ^{add4}
Min + Dim	6th	Cm ^{m6}
Maj + Dim	6th	C ^{m6}
Maj + Min	Minor 7th	C7 ^{#9}

Select **Simple PlayStyle** in the **Secret Chords** menu to access these chords in **Simple Play Style** mode only, or **All PlayStyle** to access these chords in all **Play Style** modes.

14.9 Quantization

Enables automatic rhythmic correction for **Loop Mode recordings**:

- **None** – No quantization.
- **Fractions** – Choose from 1/4, 1/8, 1/8t, 1/16, 1/16t or 1/32 to snap notes to the nearest subdivision.

14.10 Metronome Click

- Select different **Metronome Sounds** to match your preference.

14.11 Velocity Sense

- Toggle **Velocity** ON/OFF.
- When ON, note volume and intensity will respond to playing dynamics (how hard the key is pressed) for some sounds.

14.12 Auto Power Off

- You can set your Orchid to **Auto Power Off** after 10 or 30 minutes of inactivity, or set it to **Never** if you're playing it on a gig.

14.13 Version

- Displays the **current firmware version** running on Orchid.

14.14 Upgrade Firmware

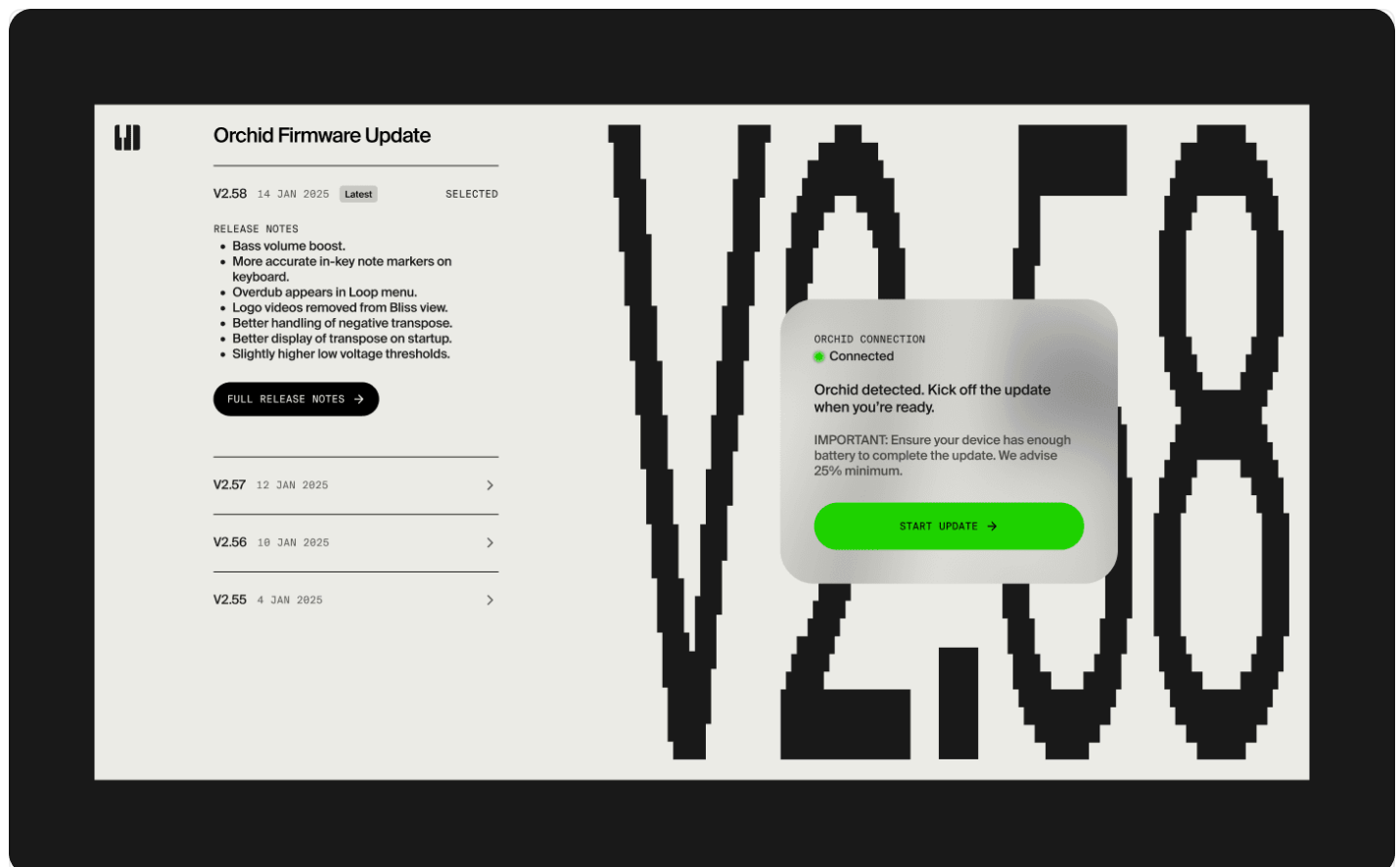
We recommend that you upgrade the firmware on your Orchid before your first use, and keep an eye out for future firmware upgrades. To update Orchid's firmware:

1. Visit firmware.telepathicinstruments.com using **Google Chrome** or **Opera** on your computer.
2. Connect Orchid to your computer via **USB-C**.
3. If prompted, allow WebMIDI to access your Orchid.
4. Navigate to the **Options** menu and select **Upgrade firmware**.
5. If an update is available, the **Start Update** button will become active.
6. Click **Start Update** to begin the process.
7. Do not navigate away from the page, close the browser tab, or disconnect the Orchid while the update is in progress.

Extension Addition
Version
Upgrade firmware

Important:

- **DO NOT INTERRUPT** the process, as it may cause issues.
- If you encounter problems, contact **Orchid Support** for assistance.



15. Synth Engine

The **Orchid Synth Engine** is a rich and powerful sound generator, comprising three distinct synthesis methods. These three synth engines have been carefully designed to provide a broad range of sonic possibilities, from warm, analogue-like tones to digital textures with intricate harmonic content.

15.1 Synth Architecture

Orchid is powered by **three distinct synthesizers**:

- **Polyphonic Virtual Analog Subtractive Synth** – Modelled after classic analogue synthesizers, featuring warm oscillators and rich filters. It produces the warm, rich, and organic sounds that are the hallmark of the 1970s and 1980s. It is ideal for creating lush pads, powerful leads, and classic keyboard tones.
- **FM Synthesizer** – Capable of producing complex harmonic textures and evolving soundscapes. It excels at creating bright, metallic, bell-like tones, as well as ethereal, otherworldly soundscapes that are impossible to achieve with subtractive synthesis alone.
- **Vintage Reed Piano Emulation** – Modelled after a legendary **60s electric piano**, including mechanical noise samples for added realism. It provides a touch of vintage soul and character to your music.

Each of these engines contributes to the overall flexibility and depth of **Orchid's sound design**. You can directly access the full power of these synthesizers using the **Pistil** plugin.

15.2 Virtual Analog & FM Synth Structure

The **Virtual Analog and FM Synthesizers** share a flexible architecture featuring:

- **4 Oscillators** per synth, providing rich sound sources with multiple waveforms
- **4 LFOs** for modulation and creating dynamic movement
- **4 Envelope Generators** for shaping amplitude, filter, and modulation over time
- **Adjustable Filter Types** for sound sculpting

15.3 Sound Selection & Presets

- **Presets are pre-designed to take full advantage** of the synth engine's capabilities.
- **Orchid does not currently allow direct editing** of synth parameters, but presets have been carefully crafted to offer **a wide range of expressive sounds**.

16. Troubleshooting & FAQs

If your **Orchid** isn't functioning as expected, refer to the troubleshooting guide below to diagnose and resolve common issues.

16.1 Common Issues & Solutions

Issues

Problem	Possible Cause	Solution
No sound output	Volume too low	Turn up the Volume Knob .
Chords sound sloppy or loose	Slop mode enabled	Check Performance Settings .
Orchid isn't playing in tune	Key Mode or voicing issue / Transpose	Check Key Mode and Voicing Settings . Also check Transpose isn't on.
Wrong chords being played	Key Mode altering accidentals	Exit Key Mode or change Key Settings .
Cannot save sound settings	Save function not used properly	The Save State Feature will be coming in an update soon.
Notes jump around octaves	Voicing Dial affecting placement	Adjust Voicing Dial or Single Notes Mode .
Bass notes not audible	Bass volume too low or incorrect mode	Increase Bass Volume and check Bass Menu .
Only bass notes playing	Bass mode settings incorrect	Check Bass Menu , ensure correct setting.
Performance is the same for each sound	Performance is locked	Long press Performance Dial to lock or unlock.
Orchid screen stuck on "Ready for Firmware" after updating	Orchid wasn't actually updating	Turn off and on again, check battery is at least 25%, make sure USB-C cable is data capable, run upgrade again.
Unexpected creative bursts	Normal function of Orchid	The Orchid is working as intended!

16.2 Frequently Asked Questions (FAQs)

Getting Started

Q: I have my Orchid. Where do I begin?

A: Charge your unit and update to the latest firmware to access the latest features. Then, start by playing a chord!

Q: Why isn't my unit turning on?

A: Hold down the **Power Button** for **10 seconds**. If that doesn't work, visit the support page for instructions on rebooting the device.

Hardware

Q: Can it charge and play at the same time?

A: Yes, however charging and the use of USB MIDI is currently not possible.

Q: Was I meant to receive a charger in the box?

A: No. **Orchid** does not include a charger to reduce electronic waste. Any **USB-C charger** will work.

Q: How do I charge Orchid?

A: Use any **USB-C charger or laptop charger**. We recommend using a fast charger. Note that a laptop charger will not work when the battery is critical or depleted.

Q: Can I connect Orchid with other hardware?

A: Yes. **Orchid** connects to:

- **Audio Equipment** via the **3.5mm line-out jack**.
- **MIDI Devices** via the **5-pin MIDI-OUT jack**.
- **Computer/DAW** via **USB-C** for MIDI communication.

Q: Can I play a chord without using the Chord Type buttons?

A: Yes, by pressing the **Key Knob**. When the **Key LED** is on, single-finger chords play in the selected key.

Software

Q: Can I create my own sounds on Orchid?

A: Yes. You can customize **Sounds** on Orchid and save them as **User Sounds**. You can create sounds from scratch using **Pistil** on your computer.

Q: Will new sounds be released?

A: Yes, via firmware updates.

Q: Why aren't my sounds velocity sensitive?

A: Ensure **Velocity Sensitivity** is enabled in **Options > Velocity Sense**. Some sounds on Orchid have yet to have velocity assigned. This will be fixed in upcoming versions.

Q: Can I save recorded loops?

A: Yes! We added the ability to save **Loops** with FIRMWARE 3.84.

Q: Can Orchid send a MIDI clock signal?

A: Yes.

Q: Can I change the speed of a recorded loop?

A: Yes. Turning the **BPM Knob** adjusts the loop's tempo.

Q: How many tracks can I layer in Loop Mode?

A: Unlimited.

Q: How do I use the bass synth?

A: Long-press the **Bass Dial** to choose a mode, short-press to turn it Off / On.

Q: Why can't I hear bass notes?

A: Increase **Bass Volume** by holding **Volume** and turning **Bass Volume** up.

Q: Why am I only hearing bass notes and not chords?

A: Long-press **Bass** to access the **Bass Menu**; you may have **Bass Single Notes** or **Solo Mode** selected.

Q: When I press an extension (6, m7, M7, 9), why don't I hear a chord?

A: When **Key Mode** is OFF, you must press a **Chord Type Button** (Dim, Min, Maj, Sus) to define the chord structure, then use extensions in combination with the **Chord Type**.

Q: Why are some of my single notes playing in a different octave from others?

A: The **Voicing Dial** affects octave placement. In **Options > Single Notes**, select **Full Octave** for minimal shifts, or **Split Mode** for broader exploration.

Support & Assistance

Q: How can I hard reboot my Orchid?

A: Hold down the **Power Button** for **10 seconds**.

Q: Can I revert to older firmware?

A: Yes. Just head to firmware.telepathicinstruments.com, select the older firmware that you would like to revert to, plug in your Orchid, and click **Start Update**.

17. Compliance & Safety

17.1 Regulatory Compliance

This product complies with the essential safety and electromagnetic compatibility standards required in various regions:

CE Compliance

This product complies with the essential requirements and other relevant provisions of **EU Directive 2014/30/EU (EMC)** and **2014/35/EU (Low Voltage)**.

FCC Compliance

This device complies with **Part 15 of the FCC Rules**. Operation is subject to the condition that this device does not cause harmful interference.

17.2 Battery & Charging

The **Orchid** contains a **rechargeable lithium-ion battery**. To ensure safe use and prolong battery life, follow these guidelines:

- **Use any standard USB-C charger.** For maximum efficiency, connect the **Orchid** to a wall socket.
- **Orchid does not charge while in USB MIDI mode.**
- **Do not overcharge.** Overcharging can shorten battery life and cause overheating.
- **Do not puncture, disassemble, or damage the battery.** If the battery swells, leaks, or emits an odor, discontinue use and contact support.
- **Store in a cool, dry place.** Avoid direct sunlight or moisture when storing for extended periods.
- **Charge to 50% before long-term storage** to maintain battery health.

17.3 Safety Warnings & Precautions

Electrical Hazard

Always disconnect **Orchid** from power before cleaning or performing maintenance.

Risk of Hearing Damage

Avoid listening at **high volumes for extended periods** to prevent potential hearing loss.

Do Not Water Your Orchid

Keep **Orchid** away from liquids to prevent electrical failure or damage.

Children's Safety

While **Orchid** is a great tool for learning music, it should only be used under **supervision** by children.

Heat Damage

Avoid prolonged exposure to **extreme heat** or **direct sunlight**, as this may cause battery overheating or failure.

18. Maintenance & Care

Proper maintenance ensures **Orchid** remains in optimal condition for years to come. Follow these care guidelines:

18.1 Cleaning

- Use a **dry, lint-free cloth** to clean the surface of the Orchid.
- Avoid using **harsh chemicals, including pesticides or herbicides**.

18.2 Storage

- Store the **Orchid** in a **cool, dry place**, away from **direct sunlight** and **moisture**.
- Whether it's on a studio shelf, next to other synths, or on the couch beside the remote (unlike your other synths), keep it safe from potential damage.

Note: If you're reading this in the distant future after finding an Orchid in an antique store – hello! We hope musical technology has evolved nicely since we designed this.

18.3 Software Updates

Orchid software is frequently updated to improve features and performance. Keep your **Orchid** up to date by:

- Checking **email notifications** or the **online Patreon community**.
- Visiting **firmware.telepathicinstruments.com** for the latest firmware.
- Plugging **Orchid** into your computer via **USB-C** and following the update instructions.